

Kazi Tasnim Zinat

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PhD Researcher with experience in sequential & time-series data analysis (healthcare, clickstream, maintenance logs) using machine learning and visual analytics. Designed LLM-powered systems for structured knowledge extraction, human-in-the-loop data analysis and implemented neural network architectures for causal inference. Developed domain-specific benchmarks to evaluate Multimodal Language Models. Experienced with Python, PyTorch, AutoGen, React, D3.js, scikit-learn, LlamaIndex, Linux and HPC cluster.

EDUCATION

University of Maryland, College Park

*Ph.D. in Computer Science; (Advisor: Leo Zhicheng Liu)
Master of Science in Computer Science; GPA: 3.97/4*

Aug 2019 – Apr 2026

College Park, MD

Bangladesh University of Engineering & Technology

Bachelor of Science in Computer Science and Engineering; GPA: 3.71/4

Feb 2013 – Sep 2017

Dhaka, Bangladesh

EXPERIENCE

Applied Science Research Intern - Amazon Web Services

Bedrock Generative AI Team

May 2023 - August 2023

Santa Clara, CA

- Engineered a **hybrid Transformer-CNN architecture** on AWS SageMaker using **multi-headed self-attention** and **multi-objective** training (NLL, cross-entropy) to capture long-range and local temporal patterns.
- Reduced prediction error by 88% (RMSE)** on industrial maintenance logs; improved simulation performance by ~50% (NLL); and identified treatment-effect shifts aligned with **real-world healthcare logs**, with findings published at ICANN 2025.

Applied Science Research Intern - Amazon Web Services

Machine Learning Solutions Lab

May 2022 - August 2022

Santa Clara, CA

- Developed a causal inference framework** extending **Rubin's model** to temporal point processes, formalizing treatment-effect estimation for non-i.i.d. event sequence data.
- Implemented counterfactual estimators** with intervention-aware **propensity score matching** and validated framework robustness through systematic ablation studies of diverse intervention types.

Graduate Research Assistant - Human Computer Interaction

University of Maryland (Human-Data Interaction Lab)

Spring 2021 – Present

College Park, MD

- Created **ProcVQA**, a first-of-its-kind benchmark for Vision-Language Models (**VLMs**), evaluating 21 frontier models (e.g., GPT-4, Gemini) on sequence, graph and tree-based reasoning across 118 process visualizations.
- Engineered zero-shot evaluation pipelines using **GCP** and **Hugging Face** to quantify **hallucination rates** and document systematic instruction-following failures in high-density visualizations.
- Developing a **RAG**-based Knowledge Base with **Docling** & **LlamaIndex** to process 100+ research papers; implemented multi-modal figure indexing and hybrid retrieval (vector + BM25) for structured data extraction.
- Delivered **source-attributed recommendations** with inline citations via a **FastAPI** backend and **React** chat interface, enabling discovery for non-technical users.
- Architected a **Multi-Agent System** using **AutoGen** to decompose natural language research goals into executable code; utilizing a systematic workflow (user goal → planning/decomposition → framework mapping → technique selection → code execution) with human-in-the-loop approvals.

Graduate Research Assistant - Bioinformatics

University of Maryland (HCBravo Lab)

Fall 2019 – Summer 2020

College Park, MD

- Developed **scTreeViz**, an open-source R package (BioConductor) for single-cell RNA-seq that integrates **PCA**, **hierarchical clustering**, and interactive icicle tree visualizations.
- Engineered data transformation pipelines to convert Seurat and SingleCellExperiment objects into interactive icicle tree-based visualizations, enabling discovery of complex cell-type hierarchies in high-throughput genomics data.

SELECTED PUBLICATIONS

P1. *ProcVQA: Benchmarking the Effects of Structural Properties in Mined Process Visualizations on Vision-Language Model Performance* (2025).

K.T. Zinat, S.M. Abrar, S. Saha, S. Duppala, S.N. Sakhamuri, Z. Liu. **EMNLP Findings** (ACL).

P2. *Uncovering Causal Relation Shifts in Event Sequences under Out-of-Domain Interventions* (2025).

K.T. Zinat, Y. Zhou, X. Lyu, Y. Wang, P. Xu. **ICANN** (Springer Nature Switzerland).

P3. *A Multi-Level Task Framework for Event Sequence Analysis* (2024).

K.T. Zinat, S.N. Sakhamuri, A.S. Chen, Z. Liu. **IEEE VIS** (TVCG).

P4. *A Comparative Evaluation of Visual Summarization Techniques for Event Sequences* (2023).

K.T. Zinat, J. Yang, A. Gandhi, N. Mitra, Z. Liu. **EuroVis** (Computer Graphics Forum).

P5. *Comparing Native and Non-native English Speakers' Behaviors in Collaborative Writing* (2025).

Y. Chen, Y. Xiao, K.T. Zinat, N. Yamashita, G. Gao, Z. Liu. **ACM CHI**.

W1. *Evaluating VLMs as Accessibility Bridges for Process Visualizations* (2025).

K.T. Zinat, S.M. Abrar, S. Duppala, S.N. Sakhamuri, Z. Liu. **CVPR VizWiz Grand Challenge Workshop**.

W2. *Comparing Native and Non-native English Speakers' Behaviors* (2025).

Y. Chen, Y. Xiao, K.T. Zinat, N. Yamashita, G. Gao, Z. Liu. **NAACL In2Writing Workshop**.

Full list available at [Google Scholar](#).

SELECTED PROJECTS

Behavioral Analytics for Design Recommendation | *Sequential RecSys, Python, Statistics* 2025-2026

- Defined a **behavioral taxonomy** mapping user actions to industry KPIs: **Impressions**, (**CTR**), **Dwell Time** and **Conversion**, utilizing interaction logs to model user intent from **implicit feedback signals**.
- Applied **Sequential Pattern Mining** and **n-gram analysis** to discover recurring behavioral subsequences, providing a foundation for **Session-Based Recommendation** strategies
- Performed **Cohort Analysis** on designer interaction sequences to quantify how recommendation timing modulates user **exploration-exploitation** behaviors.
- Applied statistical modeling (Kruskal-Wallis, post-hoc tests) on interaction logs, identifying strategies that correlate with higher design variety and quantity.

Dense Retrievers for Open-Domain QA | *NLP, Transformers, Hugging Face, BERT* 2021

- Analyzed a Dense Passage Retrieval (DPR) system featuring a **BERT** dual-encoder architecture for open-domain question answering on the QANTA 2018 dataset.
- Engineered a data processing pipeline to convert 100K+ quiz bowl questions into DPR format, including partial question generation for incremental answering.
- Fine-tuned models using an **in-batch negatives** training strategy with frozen passage encoder (21M Wikipedia embeddings), achieving 42.5% top-100 accuracy; analyzed performance gap against sparse TF-IDF baseline.

Visual Summarization Benchmarking Framework | *React, D3.js, Python, Statistics* 2023

- Developed an **end-to-end benchmarking platform** to evaluate three major visual summarization techniques.
- Re-implemented legacy visualization algorithms in a **modular React/D3.js stack**, ensuring algorithmic parity while optimizing for web-based performance and interactivity.
- Conducted a large-scale crowdsourced user study** involving **180 participants** and **1,620 observations** across 6 diverse datasets to quantify the accuracy-efficiency tradeoff of each technique.

MultiXplore: Visualization Recommendation System | *Recommendation, Ranking, Python, Flask, Postgres* 2019

- Co-developed **MultiXplore**, a full-stack recommendation system using **Python**, **Flask**, **PostgreSQL**, and **D3.js** to surface insights from high-dimensional datasets (5M+ tuples).
- Engineered a **Content-Based Filtering engine** that utilizes **Cosine Similarity** and **KL-Divergence** to rank and recommend high-dimensional visualizations based on statistical feature extraction.
- Implemented **automated SQL query generation** and **Relevance Feedback loop** for interactively refining recommendation rankings, addressing the **cold-start problem** through pre-computed similarity indexes.

TECHNICAL SKILLS

Languages & Databases: Python, C++, JavaScript, Java, R, SQL

Generative AI & LLMs: AutoGen, LlamaIndex, Docling, VLM, RAG, Prompt Engineering, Multimodal AI.

AI/ML Frameworks: PyTorch, Hugging Face, TensorFlow, scikit-learn

Deep Learning: Transformers, CNNs

Web Development: React, FastAPI, HTML5/CSS3, Node.js

Data Visualization: D3.js, Tableau, Plotly, Leaflet.js

Tools & Platforms: Linux, Git, Docker, AWS SageMaker, Google Cloud

GRADUATE COURSEWORK

Multimodal Foundation Models, Foundations of Deep Learning, Causal Inference and Evaluation Methods, Computational Linguistics, Information Visualization, Interactive Data Analytics, Advanced Numerical Optimization

SELECTED ACHIEVEMENTS & SERVICE

Conference Reviewer: IEEE VIS, EuroVis, ACM CHI, PacificVis; Special Recognition for Outstanding Reviews×3

Grants & Fellowships: AWS Cloud Credit for Research (\$4000, 2023), Jacob K. Goldhaber Travel Grant (2023),

Grace Hopper Celebration (GHC) Student Scholar (2023), Dean's Fellowship (UMD, 2019–2020)

Teaching: Served as Teaching Assistant for 7 courses, including Information Visualization, Programming Languages, Bioinformatics Algorithms, and Web Application Development with JavaScript

Mentoring: Guided 3 undergraduate and 3 masters students to conduct literature review, write research papers and develop interactive visualization applications, resulting in publications at top-tiered conferences